need a function that will get the mean field vector for the neighbors around an agent.

Listing 9.16. Calculating the mean field action

```
def get_onehot(a,l=21):                                                    1
    x = torch.zeros(21)
    x[a] = 1
    return x

def get_scalar(v):                                                         2
    return torch.argmax(v)

def get_mean_field(j,pos_list,act_list,r=7,l=21):                          3
    neighbors = get_neighbors(j,pos_list,r=r)                              4
    mean_field = torch.zeros(l)
    for k in neighbors:
        act_ = act_list[k]
        act = get_onehot(act_)
        mean_field += act
    tot = mean_field.sum()
    mean_field = mean_field / tot if tot > 0 else mean_field               5
    return mean_field
```

- ***1*** Converts integer representation of action into one-hot vector representation
- ***2*** Converts one-hot vector action into integer representation
- ***3*** Gets the mean field action of agent j; pos_list is what is returned by env.get_pos(team1), and l is action space dimension
- ***4*** Finds all the neighbors of the agents using pos_list
- ***5*** Makes sure we don’t divide by zero

The `get_mean_field` function first calls the `get_neighbors` function to get the coordinates of all the agents for agent j. The `get_mean_field` function then uses these coordinates to get the agents’ action vectors, add them, and divide by the total number of agents to normalize. The `get_mean_field` function expects the corresponding action vector `act_list` (a list of integer-based actions) where indices in `pos_list` and `act_list` match to the same agent. The parameter `r` refers to the radius in grid squares around agent j that we want to include as neighbors, and `l` is the size of the action space, which is 21.